Data Mining and Data Warehousing 3

Data Visualization & Data Similarity and DIssimilarity Measurement

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Statistical Descriptions Visualization

Measuring Data Similarity and Dissimilarity

Proximity Measures for Nominal Attributes

Proximity Measure for Binary Attributes

Dissimilarity of Numeric Attributes

Proximity Measure for Ordinal Attributes

Dissimilarity for Attributes of Mixed Types

Cosine Similarity

Statistical Descriptions Visualization

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Boxplot

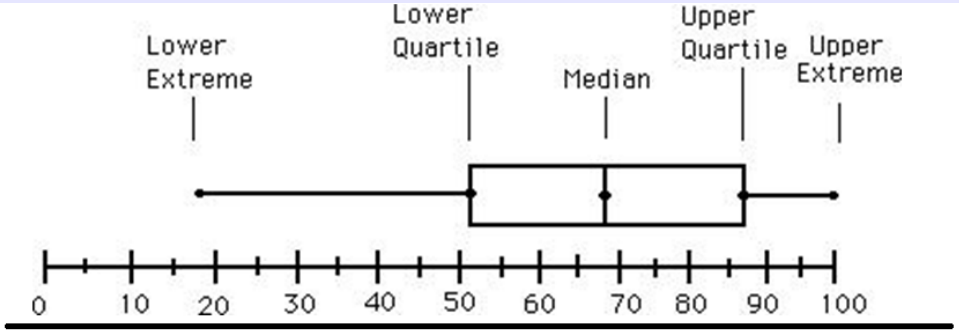
- Boxplots are a popular way of visualizing a distribution
- A boxplot incorporates the five-number summary as:

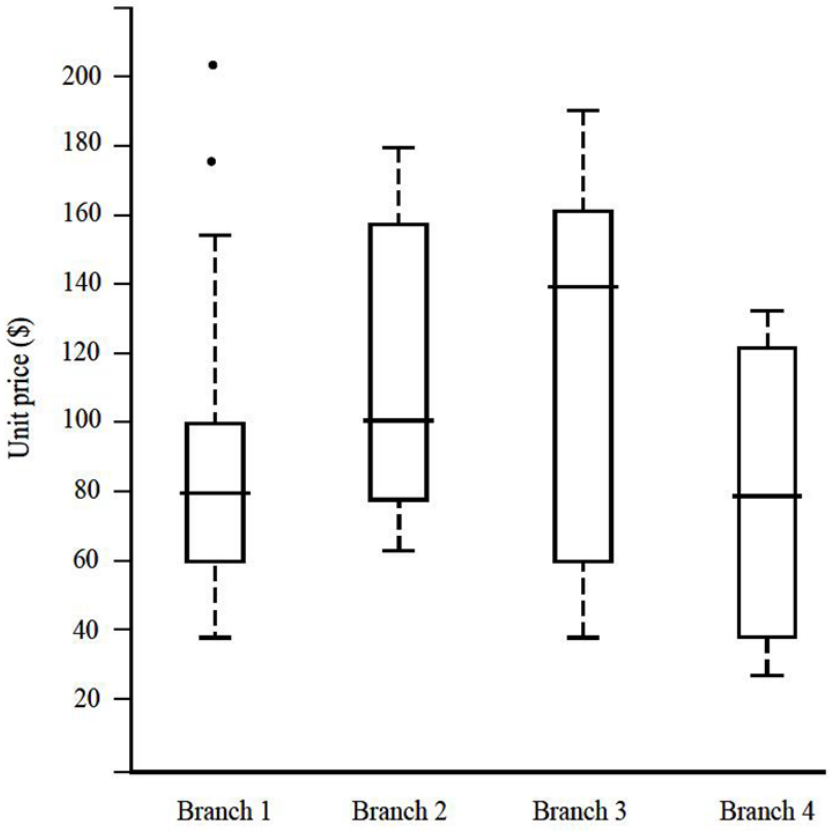
  Minimum, Q1, Median, Q3, Maximum

  - Data is represented with a box
  - The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
  - The median is marked by a line within the box
  - Two lines (called whiskers) outside the box extend to the smallest (Minimum) and largest (Maximum) observations
  - Points beyond a specified outlier threshold, plotted individually

Statistical Descriptions Visualization...

Boxplot...



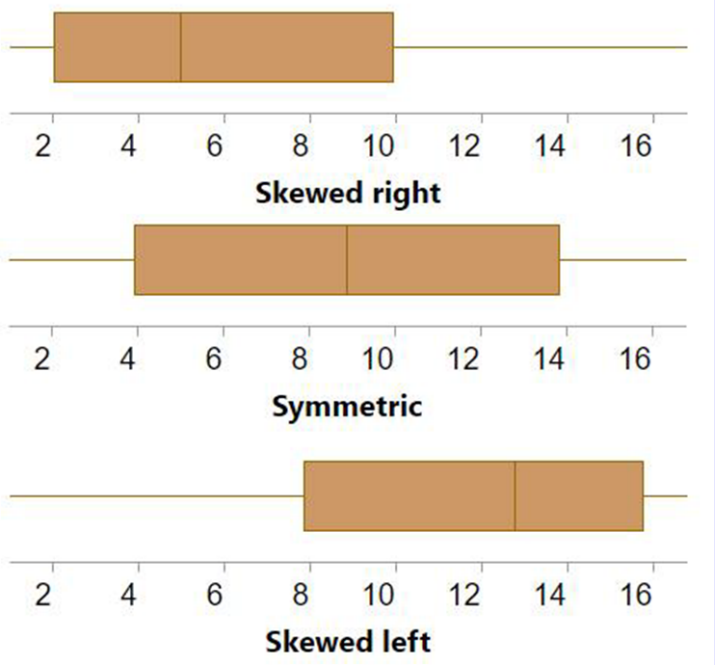


Ex: Boxplot for the unit price data for items sold at four branches of AllElectronics during a given time period

Statistical Descriptions Visualization...

Boxplot...

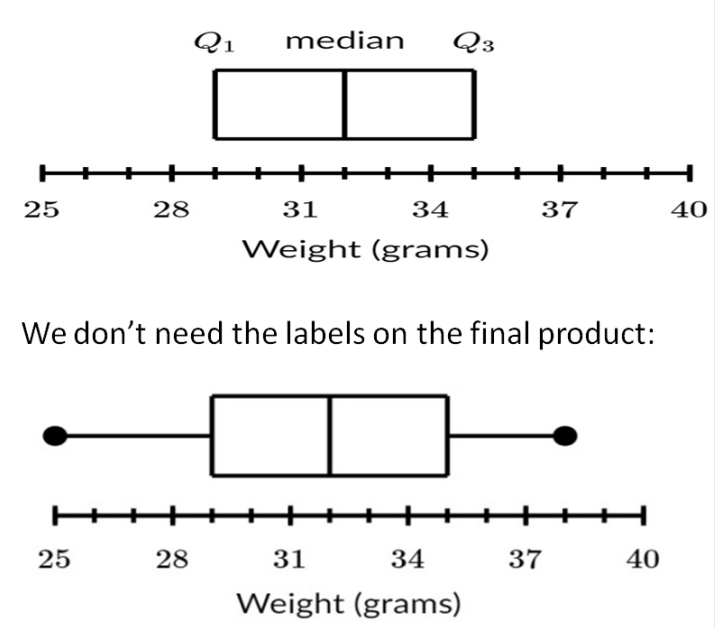
- Boxplots often provide information about the shape of a data set
- If most of the observations are concentrated on the low end of the scale, the distribution is skewed right; and vice versa
- If a distribution is symmetric, the observations will be evenly split at the median



Statistical Descriptions Visualization...

Boxplot...

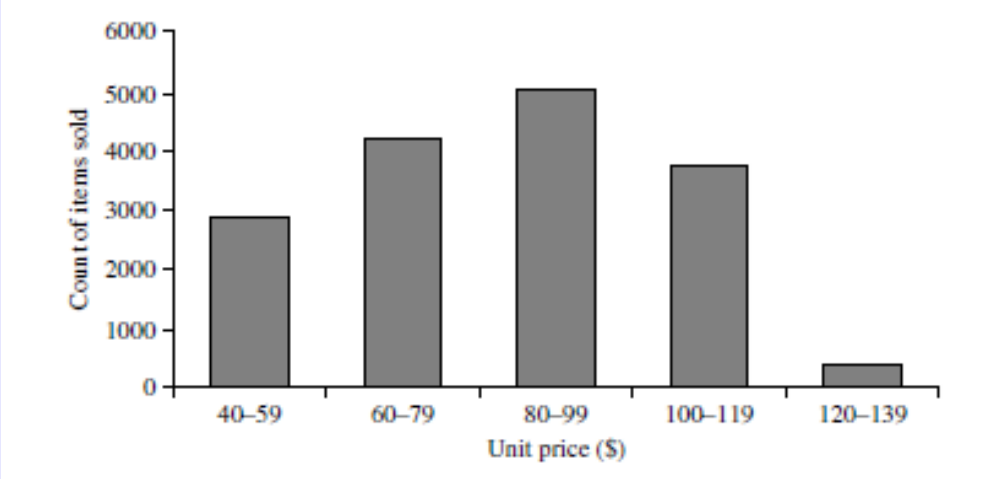
- Ex: 25, 28, 29, 29, 30, 34, 35, 35, 37, 38
- Median -> 32
- First quartile is the median of the data points to the left of the median: 25, 28, 29, 29, 30. So, Q1->29
- Third quartile is the median of the data points to the right of the median: 34, 35, 35, 37, 38. So, Q3->35
- Min-> 25 and Max->38



Statistical Descriptions Visualization...

Histogram Analysis

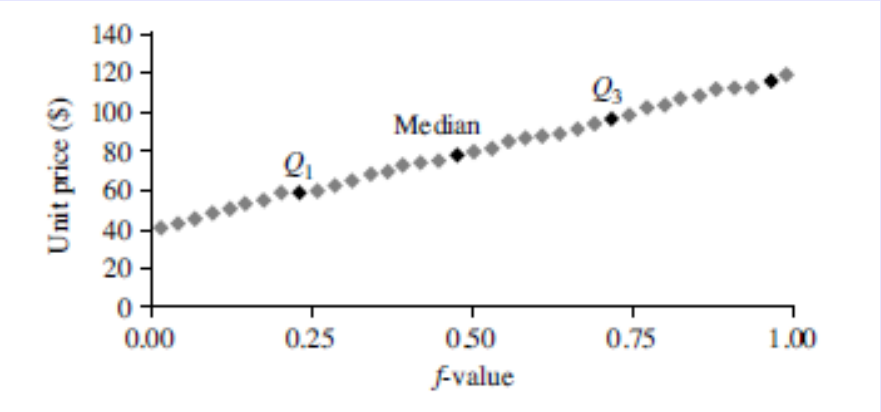
- Plotting histograms is a graphical method for summarizing the distribution of a given attribute X. The height of the bar indicates the frequency (or count) of that X value
- Differs from a bar chart in that it is the area of the bar that denotes the value, not the height as in bar charts, a crucial distinction when the categories are not of uniform width
- The categories are usually specified as non-overlapping intervals of some variable. The categories (bars) must be adjacent



Statistical Descriptions Visualization...

Quantile Plot

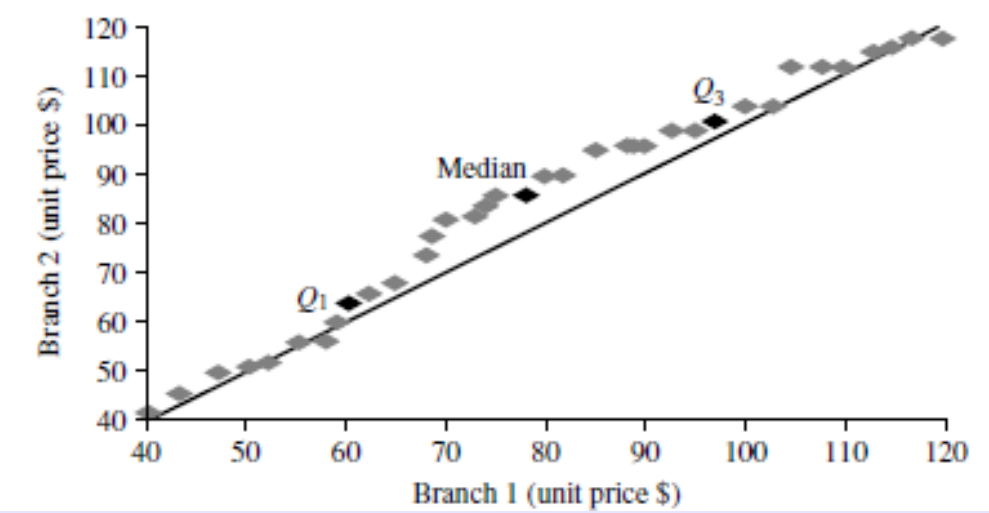
- It displays all of the data for the given attribute (both behavior and occurrences)
- It plots quantile information
- For a data x_i data sorted in increasing order, f_i indicates that approximately $100 * f_i\%$ of the data are below or equal to the value x_i
  where $f_i = \frac{i-0.5}{N}$
- It compares different distributions based on their quantiles
- It allows the user to view whether there is a shift in going from one distribution to another



Statistical Descriptions Visualization...

Quantile-Quantile Plot

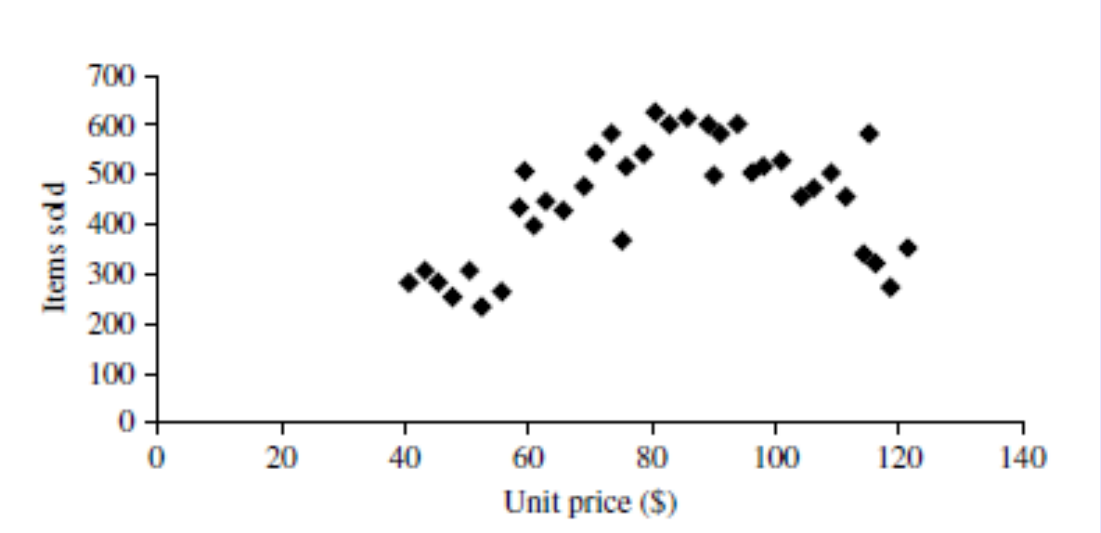
- It graphs the quantiles of one univariate distribution against the corresponding quantiles of another
- It is a powerful visualization tool in that it allows the user to view whether there is a shift in going fromone distribution to another
- Ex: unit price of items sold at Branch 1 vs. Branch 2 for each quantile. Unit prices of items sold at Branch 1 tend to be lower than those at Branch 2



Statistical Descriptions Visualization...

Scatter Plot

- Scatter plot is a type of data visualization that shows the relationship between different variables
- This data is shown by placing various data points between X-axis and Y-axis
- Provides a first look at bivariate data to see clusters of points, outliers, corelation etc
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane



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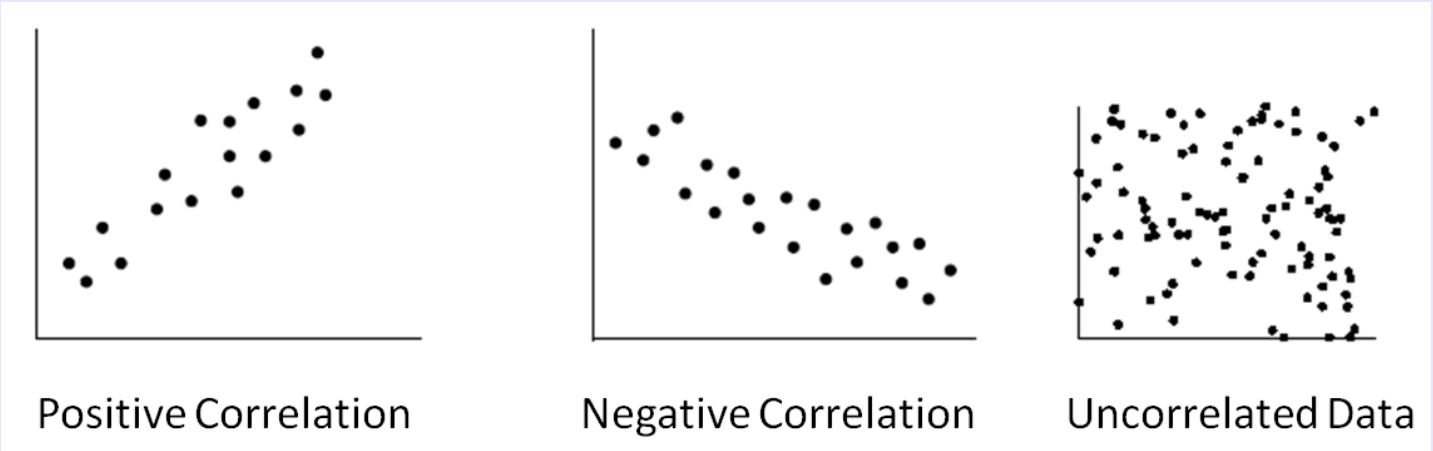
Scatter Plot with Data Correlation

- Two attributes, X, and Y, are correlated if one attribute implies the other
- Correlations can be positive, negative, or null (uncorrelated)
  - If the plotted points pattern slopes from lower left to upper right, this means that the values of X increase as the values of Y increase, suggesting a positive correlation
  - If the pattern of plotted points slopes from upper left to lower right, the values of X increase as the values of Y decrease, suggesting a negative correlation



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Measuring Data Similarity and Dissimilarity

[redacted]

- In data mining applications such as clustering, outlier analysis, classification it is required to find how much similar or dissimilar one data is in comparison to the other

- **Similarity**
  - A Numerical measure of how alike two data objects are
  - Value is higher when objects are more alike and lower when objects are more dissimilar
  - Often falls in the range [0,1]

- **Dissimilarity**
  - A Numerical measure of how different two data objects are
  - Value is higher when objects are more dissimilar and lower when objects are more alike
  - Minimum dissimilarity is often 0 and the upper limit varies

- Proximity refers to a similarity or dissimilarity

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Data Matrix and Dissimilarity Matrix

- **Data Matrix**
  - object-by-attribute structure
  - It stores the n data objects in the form of a relational table, or n-by-p matrix (n objects x p attributes)
  - Each row corresponds to an object
  - It is a *two-mode* matrix as it is made up of rows and columns
- **Dissimilarity Matrix**
  - object-by-object structure
  - It stores a collection of proximities that are available for all pairs of n objects. It is often represented by an n-by-n table
  - It is a *one-mode* matrix as it is made up of dissimilarity

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}.$$

Data Matrix

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \cdots & \cdots & 0 \end{bmatrix},$$

Dissimilarity Matrix

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Similarity/Dissimilarity for Simple Attributes

Similarity and dissimilarity between two objects, x and y w.r.t. single, simple attribute

Attribute Type	**Dissimilarity**	**Similarity**
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = \lvert x - y \rvert / (n-1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = \lvert x - y \rvert$	$s = -d$, $s = \frac{1}{1+d}$, $s = e^{-d}$, $s = 1 - \frac{d - min_d}{max_d - min_d}$

Proximity Measures for Nominal Attributes

Proximity Measures for Nominal Attributes

- It can take on two or more states
- map_color is nominal with state <red, yellow, green, pink, blue>

- **Dissimilarity between Nominal Attributes**: simple ratio of mismatches
  $d(i,j) = \frac{p-m}{p}$

  where,
  m-> number of matches
  p->total number of variables / attributes

- **Similarity between Nominal Attributes**: simple ratio of matches
  $sim(i,j) = 1 - d(i,j) = \frac{m}{p}$

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Proximity Measures for Nominal Attributes

Proximity Measures for Nominal Attributes

Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

Dissimilarity Matrix: d(i,j)

$$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ d(3,1) & d(3,2) & 0 & \\ d(4,1) & d(4,2) & d(4,3) & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

Similarity Matrix: sim(i, j)= 1-d(i,j)

$$\begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

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Proximity Measure for Binary Attributes

Proximity Measure for Binary Attributes

- Binary attribute has only two states: 0 and 1
- Contingency table is used to describe the distance between two objects (let p & q) with only binary attributes
- Compute dissimilarity/similarity using the quantiles:
  - f01 -> the number of attributes where p was 0 and q was 1
  - f10 -> the number of attributes where p was 1 and q was 0
  - f00 -> the number of attributes where p was 0 and q was 0
  - f11 -> the number of attributes where p was 1 and q was 1

	Object q			
Object p		1	0	sum
	1	f11	f10	f11+f10
	0	f01	f00	f01+f00
	Sum	f11+f01	f10+f00	

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Proximity Measure for Binary Attributes...

- **DIssimilarity measure of Binary attributes**
  - **Symmetric binary variables**
    $\frac{f10+f01}{f11+f10+f01+f00}$
  - **Asymmetric binary variables**
    $\frac{f10+f01}{f11+f10+f01}$
- **Similarity measure of Binary attributes**
  - **Symmetric binary variables [Simple matching coefficients]**
    $\frac{f11+f00}{f11+f10+f01+f00}$
  - **Asymmetric binary variables [Jaccard index similarity]**
    $\frac{f11}{f11+f10+f01}$

Proximity Measure for Binary Attributes...

Proximity Measure for Binary Attributes...

Dissimilarity between binary variables

- Gender is a symmetric attribute and the remaining attributes are asymmetric binary
- Let the values Y and P be 1, and the value N be 0
- DIssimilarity in terms of only asymmetric attribute= $\frac{f10+f01}{f11+f10+f01}$

$$d(Jack, Mary) = \frac{0+1}{2+0+1} = 0.33$$
$$d(Jack, Jim) = \frac{1+1}{1+1+1} = 0.67$$
$$d(Jim, Mary) = \frac{1+2}{1+1+2} = 0.75$$

- Among three patients, Jack and Mary have similar diseases [as their distance is less]

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

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Proximity Measure for Binary Attributes...

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- Similary in terms of only asymmetric attribute [**Jaccard index similarity**] = $\frac{f11}{f11+f10+f01}$

$$d(Jack, Mary) = \frac{2}{2+0+1} = 0.66$$
$$d(Jack, Jim) = \frac{1}{1+1+1} = 0.33$$
$$d(Jim, Mary) = \frac{1}{1+1+2} = 0.25$$

- Among patients, Jack and Mary have similar diseases [as their similarity is more]

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	1	0	1	0	0	0
Mary	F	1	0	1	0	1	0
Jim	M	1	1	0	0	0	0

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Commonly used distance measures for computing dissimilarity are:

- **Euclidean Distance**
  $d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$
  where,
  n->number of dimensions (attributes)
  x_i and y_i are the ith attributes of data objects x and y

- **Manhattan Distance**
  $d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$

- **Minkowski Distance**: is a generalization of the Euclidean and Manhattan distances. It is defined as
  $(\sum_{i=1}^{n} |x_i - y_i|^p)^{\frac{1}{p}}$
  When p=1 -> it is Manhattan
  When p=2 -> it is Euclidean

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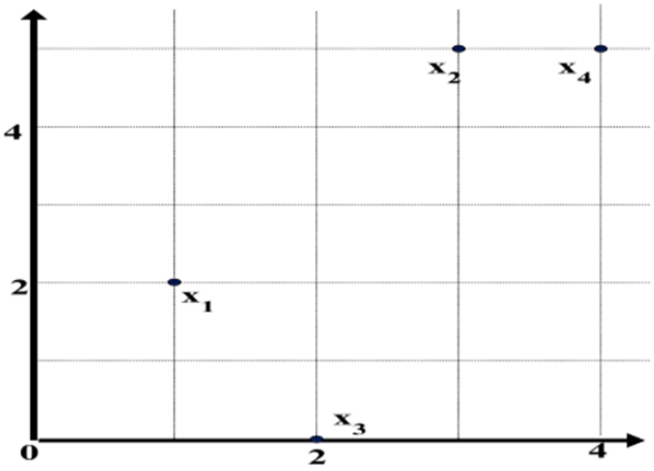

Point	Attribute1	Attribute2
x1	1	2
x2	3	5
x3	2	0
x4	4	5

Dissimilarity Matrix (Euclidean Distance)

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	5.1	5.1	0	
x4	4.24	1	5.39	0

Dissimilarity Matrix (Manhattan Distance)

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

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Point	Attribute1	Attribute2
x1	1	2
x2	3	5
x3	2	0
x4	4	5

Minkowski Distance $\left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{\frac{1}{p}}$

Where p-> real number >=1

Minkowski Distance with p value 1

Manhattan Distance $d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$

	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Minkowski Distance with p value 2

Euclidean Distance $d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$

	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	5.1	5.1	0	
x4	4.24	1	5.39	0

Proximity Measure for Ordinal Attributes

Proximity Measure for Ordinal Attributes

- Ordinal variables are discrete or continuous with meaningful order or ranking about them but the magnitude between successive ranks is unknown
- Ex: Attribute Size: <small, medium, large>

  Attribute temperature: <-30 to -10: cold, -10 to 10: Moderate, 10 to 30: warm>
- For an attribute f, let M represent the total number of possible states. Then the order/rank states is $< 1 ... M_f >$
- For an object i, let r_{if} is the rank of the ith object: $r_{if} \in 1, ..., M_f$
- Since any ordinal attribute can have a different number of states, it is necessary to normalize the range onto [0.0 to 1], where each of the rank r_{if} will be normalized to z_{if}: $z_{if} = \frac{r_{if}-1}{M_f-1}$ and then find it's dissimilarity matrix

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Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

Obj Id	Test-2 (Ordinal)	Rank (r_{ij})	Normalized Rank (z_{ij})
1	Excellent	3	1
2	Fair	1	0
3	Good	2	0.5
4	Excellent	3	1

Dissimilarity Matrix: d(i, j)

$$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ d(3,1) & d(3,2) & 0 & \\ d(4,1) & d(4,2) & d(4,3) & 0 \end{bmatrix} = \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1 & 0.5 & 0 \end{bmatrix}$$

Similarity Matrix: sim(i, j)= 1-d(i,j)

$$\begin{bmatrix} 1-0 & & & \\ 1-1 & 1-0 & & \\ 1-.5 & 1-.5 & 1-0 & \\ 1-0 & 1-1 & 1-.5 & 1-0 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0.5 & 0.5 & 1 & \\ 1 & 0 & 0.5 & 1 \end{bmatrix}$$

Dissimilarity for Attributes of Mixed Types

Dissimilarity for Attributes of Mixed Types

- The real database mostly contains mixed types of attributes. The simple approach is to process all types of attributes together by performing a single analysis to obtain a single dissimilarity matrix with a common scale of an interval [0.0, 1.0] (normalized form)
- Let the data set contains p attributes of mixed type. The dissimilarity d(i, j) between object i and object j is:
  $d(i,j) = \frac{\sum_{f=1}^{p} \delta_{ij}(f) d_{ij}(f)}{\sum_{f=1}^{p} \delta_{ij}(f)}$

  where, d_{ij} -> is the dissimilarity between object i and j
  $\delta_{ij}(f)$ -> is the indicator
  $\delta_{ij}(f)$ =0, if x_{if}/x_{jf} is missing or $x_{if} = x_{jf} = 0$ for asymmetric binary attribute
  $\delta_{ij}(f)$ =1, otherwise

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$$d(i, j) = \begin{cases} \text{if f is numeric:} & d_{ij}^{(f)} = \dfrac{|x_{if} - x_{jf}|}{max(h) - min(h)} \text{ where h is the attribute set} \\ \text{if f is nominal or binary:} & d_{ij}^{(f)} = 0 \text{ if } x_{if} = x_{jf} \text{ otherwise } d_{ij}^{(f)} = 1 \\ \text{if f is ordinal:} & \text{Compute rank } r_{if} \text{ and } z_{if} \text{ where } z_{if} = \dfrac{r_{if} - 1}{M_f - 1} \end{cases}$$

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Dissimilarity for Attributes of Mixed Types

Cosine Similarity

Dissimilarity for Attributes of Mixed Types...

Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

Dissimilarity Matrix of Nominal Attribute (test-1)

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Dissimilarity Matrix of Ordinal Attribute (test-2)

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1 & 0.5 & 0 \end{bmatrix}$$

Dissimilarity Matrix of Numeric Attribute (test-3)

max(h)=64 and min(h)=22

max(h) - min(h)= 42

$$\frac{|x_{if}-x_{jf}|}{max(h)-min(h)}=\frac{x_{i3}-x_{j3}}{42}$$

$$\begin{bmatrix} 0 & & & \\ 0.55 & 0 & & \\ 0.45 & 1 & 0 & \\ 0.4 & 0.14 & 0.8 & 0 \end{bmatrix}$$

Dissimilarity for Attributes of Mixed Types...

Statistical Descriptions Visualization

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Dissimilarity for Attributes of Mixed Types...

Using the generalized formula of mixed Attribute Disimmilarity:

$$d(i,j)=\frac{\sum_{f=1}^{p}\delta_{ij}^{(f)}d_{ij}^{(f)}}{\sum_{f=1}^{p}\delta_{ij}^{(f)}}$$

$$\begin{bmatrix} 0 & & & \\ \frac{1+1+.55}{3} & 0 & & \\ \frac{1+.5+.45}{3} & \frac{1+.5+1}{3} & 0 & \\ \frac{0+0+.4}{3} & \frac{1+1+.14}{3} & \frac{1+.5+.86}{3} & 0 \end{bmatrix} = \begin{bmatrix} 0 & & & \\ 0.85 & 0 & & \\ 0.65 & 0.83 & 0 & \\ 0.13 & 0.71 & 0.79 & 0 \end{bmatrix}$$

Cosine Similarity

Cosine Similarity

- It is a measure of similarity that can be used to compare documents or, say, give a ranking of documents with respect to a given vector of query words
- Let x and y be two vectors for comparison. Cosine similarity = $sim(x, y) = \frac{x.y}{||x||||y||}$

  where, . -> vector dot product
  ||x|| -> Euclidean norm of vector x
  ||y|| -> Euclidean norm of vector y
- The measure computes the cosine of the angle between vectors x and y
  - Cosine value of 0 means the two vectors are at 90 degrees to each other and have no match
  - The closer the cosine value to 1, the smaller the angle and the greater the match between vectors

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Cosine Similarity

Cosine Similarity...

- d1 = 3, 2, 0, 5, 0, 0, 0, 2, 0, 0
  d2 = 1, 0, 0, 0, 0, 0, 0, 1, 0, 2
- $d1.d2 = 3*1+2*0+0*0+5*0+0*0+0*0+0*0+2*1+0*0+0*2 = 5$
- $||d1|| = (3*3+2*2+0*0+5*5+0*0+0*0+0*0+0*0+2*2+0*0+0*0)^{0.5} = (42)^{0.5} = 6.481$
- $||d2|| = (1*1+0*0+0*0+0*0+0*0+0*0+0*0+0*0+1*1+0*0+2*2)^{0.5} = (6)^{0.5} = 2.245$
- $cos(d1, d2) = .3150$

Ref: J. Han, J. Pei and H. Tong, "Data Mining: Concepts and Techniques", Morgan Kaufmann, 4th edition